

Set Up an Open Virtual Switch



Open Virtual Switch or Open vSwitch supports flows, VLANs, trunking and port aggregation just like other major league switches. Now, discover how to set up an Open vSwitch on an Ubuntu 12.04 system.

Open Virtual Switch (OVS) is an open source Layer 2 virtual switch, which offers many Layer 2 features and even supports some Layer 3 protocols. It is one of the key components behind many public and private cloud offerings.

As with any switch, OVS also has two paths—a fast data path and a decision-making control path.

Control path: This is in user space and creates a table with decisions.

Data path: This is in the kernel space; packets flow through this path based on decisions made by the control path.

Working

Whenever a packet arrives, the data path checks for the existing flows, and if there are no matching flows, the packet will be sent to the user space controller along with a flow key to generate the flow. The flow key describes a packet in general terms.

Sample flow key

```
in_port(1), eth(src=e0:19:2f:10:0a:ba,
dst=00:21:f3:f1:89:a1),
eth_type(0x0800), ipv4(src=10.168.122.3 dst=10.168.122.4,
proto=17, tos=0, frag=no), tcp(src=31639, dst=8080)
```

Now, based on the flow key, the controller will either neglect or generate a new flow to the kernel space. This flow will remain in the kernel space for the next 5 seconds, and if there is no matching flow within that time span, it will expire.

Figure description

As represented by the ----- line, the first packet will go through the controller and all subsequent packets will go through the data path.